Ashutosh "Ash" Panpalia

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Career Objective: Result driven Robotics engineer specialized in full stack development for robot arm manipulation and gripper dexterous manipulation. Leveraging 5 years of engineering experience to drive innovation in robotics.

Skills:

Robotics Software: Python, ROS 2, PyBullet, Nvidia Isaac Sim, Isaac Lab, Gazebo, RViz, Nav2, SLAM, OpenCV
Robotic Learning: RL, Sim-to-Real, Diffusion Policy, Imitation Learning, Reward Shaping, Domain Randomization, PPO, DDPG, PyTorch, YOLO, SAM, FoundationPose
Control system: PID, Classical Control, Path Planning, Inverse Kinematics, Closed-Loop Control, Sensor Fusion
Perception& Sensing: Stereo Vision, Point Clouds, Visual Servoing, 3D Object Detection, 6DoF Pose Estimation
Equipment Expertise: Robotic Arms, LiDAR, Tactile Sensors, IMU, Intel RealSense, High-Speed Cameras, GPS

Work Experience:

Robotics Engineer, Zoho Corporation US *Oct 2024 - current*

- Built a full-stack robot manipulation pipeline integrating 6DoF object pose estimation with visual servoing using a wrist-mounted depth camera to achieve on-conveyor assembly/bolting task.
- Developed RL-based control pipelines in PyBullet, using reward-shaping techniques to train pick-and-place policies that achieved 83% sim2real transfer success on a Trossen Robotics 6-DOF arm.
- Developed an internal tool to reduce real image data collection and annotation effort by 95% using CAD-based synthetic data while achieving similar object detection performance.
- Set up the Isaac Lab locomotion environment for Unitree Go2 quadrapped robot, defining observation/action spaces and physics-accurate walking tasks.
- Designed the hardware and sensor architecture for a UMI-style imitation learning data collection system.

Research Assistant, EMNM Lab - University at Buffalo PI: Dr Jun Liu *Sept 2023 - Aug 2024*

- Developed closed-loop position control system for a 2-finger adaptive gripper integrating tactile sensors for real-time slip detection and contact-aware grasping.
- Integrated motors, tactile sensors, force sensor and data acquisition system for control system deployment.
- Characterized tactile sensor performance by designing experiments to determine optimal sensor sampling frequency for reliable slip detection.

Research Engineer, Machine Tool Dynamics Lab, IIT Kanpur PI: Dr Mohit Law *Feb 2023-July 2023*

- Developed experimental framework to conduct system identification of 6-DOF industrial robotic arm.
- Executed 27 pilot experiments for dynamic system identification by employing modal analysis techniques.
- Utilized insights from modal analysis of robotic arm to design robust control system for improved path tracking.

Assistant Manager QA Hero MotoCorp Ltd. *Aug 2020-Jan 2023*

- Implemented AI-powered vehicle inspection using perception model on production lines to enhance quality..
- Resolved a critical 0.1mm fitment issue on export accessories through tolerance stack-up analysis.
- Integrated cameras and ultrasonic sensors on 4 conveyors with collaboration of all teams to enhance traceability, mitigate workplace hazards, and alleviate human fatigue

Relevant Projects:

Sim-to-Sim RL Policy Transfer (Isaac Lab to Isaac Sim) *2025*

- Deployed a locomotion policy trained in Isaac Lab into Isaac Sim, reconstructing the environment and mapping observation/action interfaces to validate policy execution.

Autonomous Vehicle, Planning and Control *2024*

- Developed ROS-based motion planning algorithms using Pure pursuit and RRT for autonomous vehicle navigation.
- Implemented PID controller for steering in autonomous vehicles to achieve robust vehicle trajectory control.

Modular Robotic Arm Development for Martian Rover *2019*

- Implemented control system strategy for 5-DOF robotic arm using inverse kinematics for motion control.
- Engineered 5-DOF robotic arm, with lifting capacity of 3 Kg and reach of 1m utilizing first principle design approach.

Education

M.S: Robotics Engineering, University at Buffalo, SUNY, New York, USA (GPA: 3.78/4)

B.Tech: Mechanical Engineering, Delhi Technological University, New Delhi, India (CGPA: 8.85/10)

Leadership Experience

- Currently leading coordination between USA and India Zoho R&D robotics teams to streamline multiple projects.
- Mentored 35 underprivileged students in India for 200+hrs covering basics of Python coding, virtual mentoring.